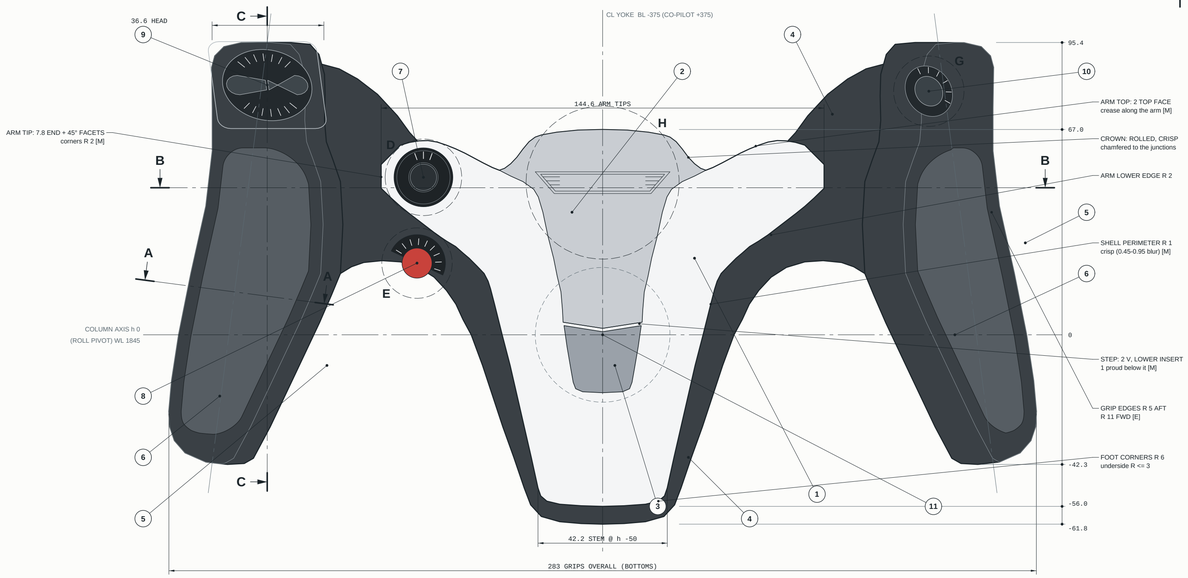
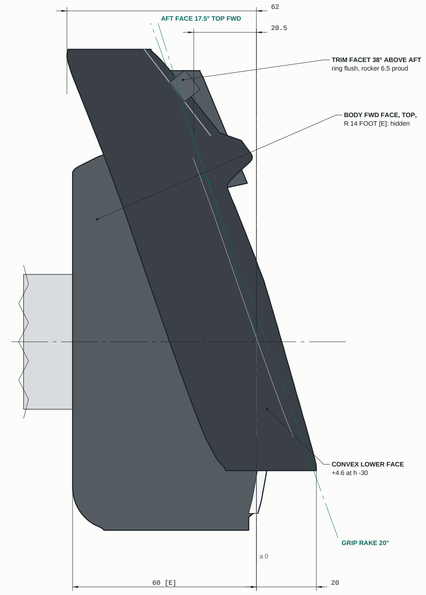


FRONT ELEVATION - PILOT YOKE

LOOKING FORWARD, AS THE PILOT SEES IT (CO-PILOT YOKE = MIRROR IMAGE) - SCALE 1:1



SIDE ELEVATION

FROM OUTBOARD, NOSE LEFT, NEUTRAL PITCH; a 0 = X 3700; COLUMN BROKEN (DETAIL J) - 1:1

ITEMS & FINISHES (render/lookdev names; 'new' = to add)

#	PART	MATERIAL	FINISH / COLOUR [SOURCE]
1	WHITE COVER (Y SHELL)	yoke_white	gloss clear-coat white, neutral-cool (0.77 / 0.78 / 0.80), fine pearl; R 1 edges [M]
2	UPPER INSERT = CROWN, BADGE	yoke_silver (new)	satin aluminium 0.62, metallic 0.85, rough 0.30; badge grooves 0.5 [M]
3	LOWER INSERT	yoke_silver_dark (new)	smoke silver 0.30, metallic 0.85, rough 0.34; 0.47 x item 2 [M]
4	BLACK BODY (RIM, NECKS)	grip_black	satin black, rough 0.42 (was 0.62): crisp edge highlights [M]
5	GRIP (2), HEADS	grip_black	satin black, R 5 aft / R 11 forward edges [M / E]
6	LEATHER PAD (2)	leather_black	charcoal 0.05, rough 0.58, 1.1 pebble grain, ~1 creases, bead [M]
7	TCS BUTTON	gloss_black / black	gloss black bezel <= 1 proud, satin button recessed [M]
8	AFCS DISCONNECT	button_red (new) / black	red 0.70 / 0.035 / 0.04, clear coat 0.3, NOT emissive [M]
9	PITCH TRIM SWITCH	black / legend_white	satin black ring + rocker, tick legends white 0.80 [M]
10	PUSH BUTTON (INBOARD HEAD)	black	satin black dished collar, smooth cap, centre dimple [M]
11	COLUMN	metal_dark	satin dark metal, dia 44 [E] (never seen)
12	COLUMN SURROUND (PANEL)	pedestal_gunmetal	satin dark grey moulding, J band 23 wide [M]

MAIN DIMENSIONS

DIMENSION (mm, YOKE-LOCAL)	VALUE	SOURCE
shell (white cover) height / about h 0	123 / -56..+67	[M] P1046408 panel_408, both yokes
arm tips / stem at h 0 / at h -50	144.6 / 65.6 / 42.2	[M] sigma 1.5
insert crown above the arm junctions	13.3 (67.0 / 53.7)	[M] parting grooves
step line / lower insert bottom	h 2.1 (2 V) / -18.9	[M]
black rim at h 10 / 0 / -50 / foot	13.0 / 10.1 / 4.5 / 5.8	[M] outboard true edge
hand-slot top	h 24 +/-3	[M] both sides averaged
aft face above the step	17.5° top fwd	[M] bundle adj., +/-4
body depth / cover	60 / 3	[E] hidden in every photo
grips overall / bottoms / head tops	283 / -42.3 / 95.4	[M] (rev H 285 / -40 / 118)
grip length / cant (tops inboard)	137.7 / 15-17°	[M] (rev H 160 / 10°)
section A-A / rake / toe-out out, in	42 x 27 / 20° / 13°, 28°	[M] +/-4 / +/-5 / +/-8
pad: top / bottom / frame	h 61 / -32.5 / 4	[M] P1046408, AOPA
TCS centre / bezel / button	(58.5, 51.4) / 19.2 / 8.6	[M]
AFCS cap centre / dia x high	(60.6, 23.4) / 9.8 x 5.5	[M]
trim centre / ring / rocker	(109.5, 81.5) / 29.5 / 27 x 8	[M]
push button centre / collar / cap	(106.5, 79.5) / 17.3 / 10.2	[M] (inboard head: -u)
column dia / axis	44 [E] / h 0 = WL 1845	kept; surround reads 1821
pitch travel / roll	+/-90 [E] / +/-70 [E]	unchanged (YOKE)

KNEES vs YOKE (L6B), REV H > REV I - CRITERIA UNCHANGED

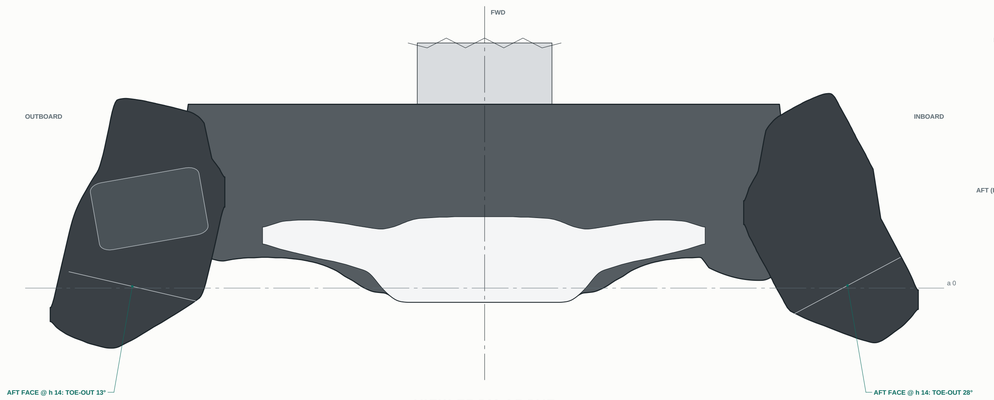
OCCUPANT	LEVEL mm	ROLL+/-70 mm	CONTACT deg	KEEP 25 deg	RUDDER mm
95th male	30 > 35	-19 > -18	17.5 > 17.5	5.0 > 2.5	-11 > -13
50th male	50 > 48	-13 > -14	25.0 > 25.0	10.0 > 10.0	5 > 3
5th male	51 > 48	-21 > -21	22.5 > 22.5	10.0 > 10.0	20 > 18
5th female	76 > 78	5 > 0	87.5 > 87.5	27.5 > 25.0	55 > 53

NOTES

1. Yoke-local mm: u outboard (pilot: -y), h up from the hub centre = the column axis / roll pivot (WL 1845, BL -/+375), a aft of the aft face at the insert step (X 3700 at neutral pitch). Drawn from model/interior.py YOKE_DESIGN rev I; L6 / L6B take the same outline (yoke_outline_yz, yoke_sweep_pieces).
2. Symmetric moulding except the switches: TCS, AFCS DISC and the trim on the OUTBOARD side, the push button on the inboard head; the co-pilot yoke is the mirror image.
3. Sources: P1046408 through the fitted camera panel_408; P1046409 / 10, the Mexico and AOPA press photos, the JTF p.18 elevation; a bundle adjustment of 4 photos for depth. All photos look from inboard: the outboard and the mirrored inboard traces are averaged; the drawn yoke, projected through the cameras, sits on the photos within 0.3-4.7 mm at the landmarks (overlays, refs/cache).
4. NO MARKINGS: the badge is an abstract winged plate, every legend a row of ticks.
5. Not modelled / unsupported: the rev H column bellows (none in 8 photos), a PTT trigger (none seen), the PC-24's extra button and grip tab.
6. Knee checks on the rev I outline: the 95th clears the yoke by 35 mm wings level (criterion 25: passes); roll to contact 17.5°, keeping 25 mm to 2.5° (rev H 5.0°) -- owner decision (open point 1 of L6B; criteria not loosened).

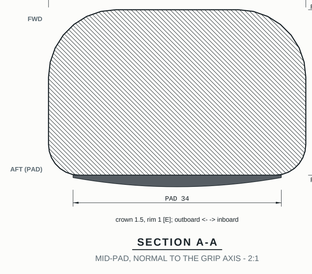
OPEN POINTS FOR THE OWNER

1. Absolute fore-aft station: the locked photos put the aft face 60 +/-30 forward of the drawn neutral (hub_dx kept: the gust-lock pose?).
2. Column: the surround's joint line reads WL 1821 +/-10 (25 under the pivot, kept); fixed J surround on the panel vs a curved arm moving with the yoke; push travel 90 [E] leaves 17 to the panel.
3. Toe-out 13 / 28 measured per grip (+/-8). symmetric 20 / 20 would suit a symmetric moulding but left the outboard pad too narrow in P1046410.
4. Hidden: body depth 60, top and foot radius, grip forward corners R 11, rocker / cap heights (+/-2).
5. PTT: none on the PRO in any photo; the PC-24 has a grip tab.



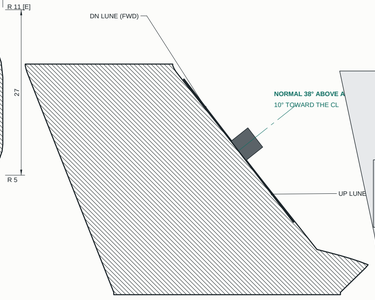
VIEW FROM ABOVE

FORWARD UP, PILOT YOKE - SCALE 1:1



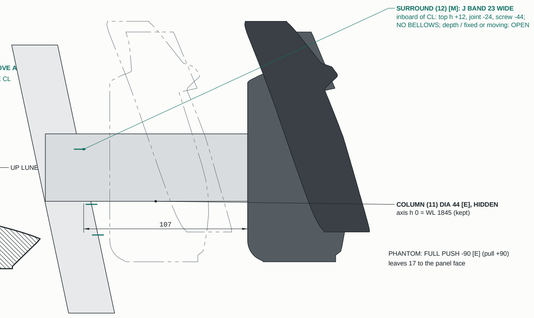
SECTION A-A

MID-PAD, NORMAL TO THE GRIP AXIS - 2:1



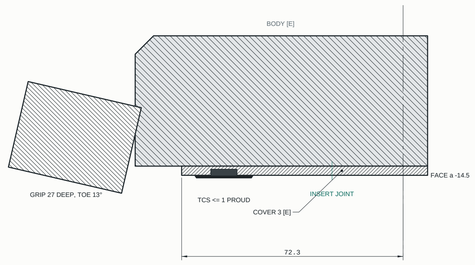
SECTION C-C (HEAD ONLY)

TRIM HEAD AT u 109.5 - 2:1



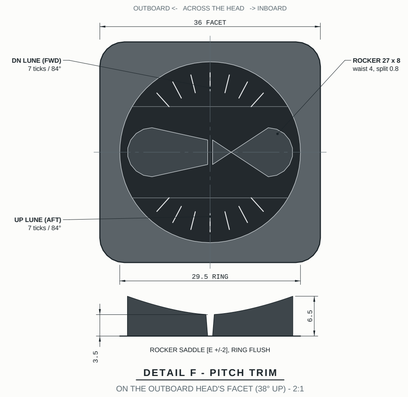
DETAIL J - COLUMN

AT THE LOWER PANEL, FROM OUTBOARD - 1:2



SECTION B-B

h 48 (BADGE / TCS), OUTBOARD HALF (THE INBOARD HALF MIRRORS IT WITHOUT THE TCS) - 1:1



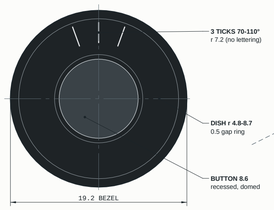
DETAIL F - PITCH TRIM

ON THE OUTBOARD HEAD'S FACET (38° UP) - 2:1

PHOTO CHECK: THE DRAWN YOKE THROUGH THE PHOTO CAMERAS (mm)

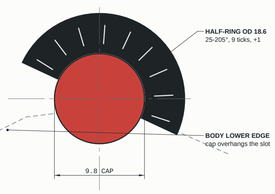
PHOTO (CAMERA)	LANDMARKS mean / max	TRACED EDGES mean (mm)
P1046408 pilot (panel_408)	2.0 / 4.0	shell 1.8, insert 1.0, badge 1.1, rim 1.8, grips 4.1, slot 4.2; neck out 2.4 / in 14 *, inboard pad 6.6 *
P1046408 co-pilot (panel_408)	2.1 / 4.2	shell 2.4, insert 1.6, badge 2.0, rim 1.7, grips 3.2, slot 4.1; neck out 2.5 / in 14 *, inboard pad 6.2 *
Mexico press photo, pilot	2.1 / 4.7	landmarks only (roll 10° in this photo)
Mexico press photo, co-pilot	2.1 / 4.3	landmarks only (roll 7°)
AOPA / Pilatus press, pilot	2.1 / 3.5	landmarks only (seen from aft-outboard)
P1046410 pilot	1.6 / 2.7	landmarks only

Only the yoke's photo pose (pitch, roll, a few mm of shift) is fitted per photo, to the landmark picks (TCS, AFCS cap, trim ring, button, badge, step ends, lower insert, foot); overlays in refs/cache/overlays/yoke (private). * traced 'as seen' from inboard: features off the trace's 4.3° plane shift toward the camera (the inboard neck and the 28° toed inboard pad); the drawing follows the 3-D fit



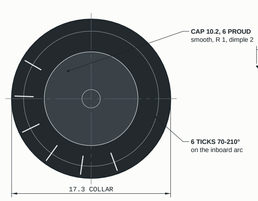
DETAIL D - TCS

ON THE WHITE ARM, u 58.5 h 51.4, <= 1 PROUD - 3:1



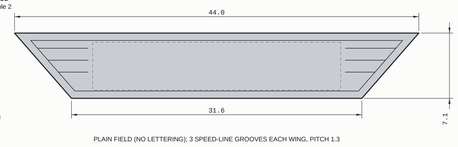
DETAIL E - AFCS DISC

u 60.6 h 23.4; CAP 9.8 x 5.5 HIGH - 3:1



DETAIL G - PUSH BUTTON

INBOARD HEAD (NOT A 4-WAY HAT) - 3:1



DETAIL H - BADGE

ABSTRACT WINGED PLATE, TOP AT h 53.2 - 3:1

REVISIONS - PC12-L6Y

REV	DATE	DESCRIPTION
A	2026-10-05	first issue (owner 2026-10-05: 'the yoke isn't quite right especially if you look at the details'): yoke re-measured on 8 PRO photos (P1046408 through the fitted camera panel_408, P1046409 / 10, Mexico and AOPA press photos, JTF p.18) incl. a bundle adjustment for depth; YOKE_DESIGN rev I replaces the rev H outline (L6 / L6B rev I)

PILATUS PC-12 PRO

PC-24-STYLE CONTROL YOKE - DETAIL

FIRST ANGLE PROJECTION

DWG NO	REV	SCALE	SHEET	UNITS
PC12-L6Y	A	AS SHOWN	A1 · 1 OF 1	mm

DRAWN	DATE
Claude - loftkit parametric model	2026-10-05

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION